

Author Statements

These are typically uploaded as one combined document or as separate fields in the submission system, depending on the journal — check the specific journal's requirements and split if needed.

CRediT Author Statement

The CRediT (Contributor Roles Taxonomy) framework is used by most Elsevier and many other journals. List every author and mark which roles they performed. Adjust to reflect actual contributions — do not leave roles unassigned or assign roles to someone who didn't perform them; this is checked and taken seriously by editors.

The author identity below is final. The role assignments remain an author-confirmation draft and must be reviewed before submission.

Role	S. B. Mahafuj Bondhon
Conceptualization	✓
Methodology	✓
Software	✓
Validation	✓
Formal analysis	✓
Investigation	✓
Data curation	✓
Writing – original draft	✓
Writing – review & editing	
Visualization	✓
Supervision	
Project administration	✓
Funding acquisition	

Conflict of Interest Statement

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

[Update this if any author has a relevant affiliation, funding relationship, or personal connection that a reader would reasonably want disclosed — e.g., if any dataset creator, cited author, or reviewer suggestion has a prior collaboration with the authors.]

Funding Statement

[State explicitly, even if the answer is "no external funding": e.g., "This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors," or, if applicable, name the specific funding source, e.g., institutional support from [BUET / Elevon TechionX / other], and any grant numbers.]

Data Availability Statement

The primary dataset used in this study is the publicly available BubbleML Pool-Boiling Subcooled FC-72 dataset (Hassan et al., BubbleML: A Multiphase Multiphysics Dataset and Benchmarks for Machine Learning, NeurIPS 2023 Datasets and Benchmarks Track), available at the official HPCForge/BubbleML repository and its associated data hosting locations. The exact source release and checksums used by this study are documented in CHECKSUMS.md; this repository does not redistribute the raw data.

Code, retained machine-readable result artifacts, configurations, and statistical-analysis scripts for this study are designated for release at:

- GitHub repository: <https://github.com/RealmeBTI/bubbleml-submission>
- Permanently archived release with DOI: [Zenodo DOI to be assigned upon initial GitHub release archival]

The repository includes a README with environment setup instructions, pinned dependency versions, and stored-result reproduction instructions. Raw data, complete checkpoints, and complete cross-condition exports remain external or unavailable as documented in `ARTIFACT_GAPS.md`; this statement does not claim a full retraining release.

Ethics Statement

[Include only if the journal requires it. This study uses only publicly available simulation data and does not involve human or animal subjects; state this explicitly if the journal's submission system requires an ethics declaration field, e.g.: "This research did not involve human participants, human data, or animals, and no ethical approval was required."]